

Registration No :

--	--	--	--	--	--	--	--	--	--

Total Number of Pages : 02

B.Tech
PCBM4302

5th Semester Back Examination 2018-19

SIGNALS AND SYSTEMS

BRANCH : AEIE, BIOMED, BIOTECH, CSE, ECE, EIE, ETC, IEE, IT, ITE

Time : 3 Hours

Max Marks : 70

Q.CODE : E544

Answer Question No.1 which is compulsory and any FIVE from the rest.

The figures in the right hand margin indicate marks.

Q1 Answer the following questions : (2 x 10)

- a) Determine whether or not the signal $x[n]$ is periodic. If $x[n]$ is periodic, determine its fundamental time period.

$$x[n] = \cos^2 \frac{\pi}{8} n$$

- b) Determine whether the signal $x[n] = 2e^{j3n}$ is energy signal, power signal nor neither.

- c) Is the system $y(t) = x(2t)$ time invariant?

- d) State and explain frequency shifting property of Fourier series.

- e) Determine the Fourier transform of unit impulse.

- f) Determine appropriate Fourier representations of the signal

$$x(t) = |\sin(2\pi t)|$$

- g) Write conditions for stability and causality of the system with rational transfer function (z) .

- h) State initial value theorem & final value theorem in Z transform.

- i) Determine the z-transform and ROC of the signal

$$x(n) = \delta[n - k], \quad k > 0$$

- j) Write down the circular time shift and circular frequency shift property of DFT.

Q2 a) The impulse response of a discrete-time LTI system is given as (5)

$$h(n) = \left(-\frac{1}{2}\right)^n u(n) + (1.01)^n u(n-1)$$

Determine whether the system is causal and/ or stable.

b) Find the convolution sum (5)

$$y[n] = (u[n+10] - 2u[n] + u[n-4]) * u[n-2]$$

Q3 a) Compute the autocorrelation of the signal $x(n) = \alpha^n u(n)$, $0 < \alpha < 1$ (5)

b) Find the FT of the signal $x(t) = \frac{d}{dt} \{ (e^{-3t} u(t)) * (e^{-t} u(t-2)) \}$ (5)

Q4 a) Find out the FS coefficients by method of inspection for the signal (5)

$$x(t) = \sin(3\pi t) + \cos(4\pi t)$$

b) Find the FT of signal $x(t) = \frac{1}{1+t^2}$ (5)

Q5 a) Suppose we are given the following information about a continuous-time periodic signal with period 3 and Fourier coefficients $X(k)$: (5)

(1) $X(k) = X(k+2)$

(2) $X(k) = X(-k)$

(3) $\int_{-0.5}^{0.5} x(t) dt = 1$

(4) $\int_{0.5}^{1.5} x(t) dt = 2$

Determine (t) .

b) Find the inverse Z-transform

(5)

$$X(z) = \frac{2 + z^{-1}}{1 - \frac{1}{2}z^{-1}}$$

With ROC $|z| > \frac{1}{2}$, using power series expansion

Q6 a) A causal system has input $x[n]$ and output $y[n]$. Use the transfer function to determine the impulse response of this system.

(5)

$$x[n] = \delta[n] + \frac{1}{4}\delta[n-1] - \frac{1}{8}\delta[n-2]$$

$$y[n] = \delta[n] - \frac{3}{4}\delta[n-1]$$

b) Use the unilateral z-transform to determine the forced response, the natural response, and the complete response of the systems described by the following difference equations with the given inputs and initial conditions.

(5)

$$y[n] - \frac{1}{3}y[n-1] = 2x[n], \quad y[-1] = 1, \quad x[n] = \left(-\frac{1}{2}\right)^n u[n]$$

Q7 Perform the circular convolution of the following two sequences $x_1(n)$ & $x_2(n)$. Illustrate the sequences of circular convolution with neat sketches. By means of the DFT and IDFT, determine the sequence $x_3(n)$ corresponding to the circular convolution of the sequences $x_1(n)$ & $x_2(n)$.

(10)

$$x_1(n) = \{2, 1, 2, 1\}, \quad x_2(n) = \{1, 2, 3, 4\}.$$



Q8 Write short answer on any TWO :

(5 x 2)

- a) Duality property of FT
- b) Causality and stability of LTI system in Z domain
- c) Properties of Convolution
- d) Parseval's Theorem